# Exact $q$-Extrapolation of Finite Sinc-Product Splines

## Recovering rational values of Rvachev's up-function at dyadic points

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A self-contained development based on the finite-spline, head–tail, derivative-tiling, and $q$-Richardson ingredients in the `ProveIt` repository

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**Abstract**

Let

$$\Phi(t) = \widehat{\mathrm{up}}(t) = \prod_{k=0}^{\infty} \operatorname{sinc}\left(\frac{\pi t}{2^k}\right), \qquad \operatorname{sinc} z = \frac{\sin z}{z},$$

and let the user-indexed finite inverse transforms be

$$Q_n(x) = \mathrm{up}_n(x) := \mathcal{F}^{-1}\left[\prod_{k=0}^{n} \operatorname{sinc}\left(\frac{\pi t}{2^k}\right)\right](x).$$

Fix a dyadic rational $x \in [-1,1]$, let $s = \mathrm{depth}_2(x)$ be the least nonnegative integer for which $2^s x \in \mathbb{Z}$, and put $d = \lfloor s/2 \rfloor$. We prove the exact eventual law

$$Q_n(x) = \mathrm{up}(x) + \sum_{m=1}^{d} C_m(x)\, 4^{-mn} \qquad (n \geq s).$$

Thus, after at most the first $s$ terms, the approximation is not merely asymptotic: it is exactly the true value plus $d$ geometric progressions, with successive ratios $4^{-1}, 4^{-2}, \ldots, 4^{-d}$. The coefficients are

$$C_m(x) = (-1)^m a_m 4^{-m} \mathrm{up}^{(2m)}(x),$$

where the positive rational numbers $a_m$ are determined by

$$\frac{1}{\Phi(z)} = \sum_{m \geq 0} a_m (2\pi z)^{2m}.$$

The termination comes from two exact facts: the omitted Fourier tail is a scaled copy of up supported inside one polynomial cell of the finite spline, and $\mathrm{up}^{(r)}(x) = 0$ at a dyadic point whenever $r > s$.

The constant term can therefore be recovered from only $d+1$ consecutive finite approximants. With $\rho = 1/4$, any $n_0 \geq s$ gives

$$\boxed{\mathrm{up}(x) = \frac{1}{(\rho;\rho)_d} \sum_{j=0}^{d} (-1)^{d-j} \rho^{\binom{d-j+1}{2}} \begin{bmatrix} d \\ j \end{bmatrix}_\rho Q_{n_0+j}(x).}$$

Equivalently, in an integer-coefficient base-4 form,

$$\boxed{\mathrm{up}(x) = \frac{\sum_{j=0}^{d} (-1)^{d-j} 4^{\binom{j+1}{2}} \begin{bmatrix} d \\ j \end{bmatrix}_4 Q_{n_0+j}(x)}{\prod_{r=1}^{d} (4^r - 1)}.}$$

These are exactly the Lagrange weights for extrapolating a polynomial from the geometric nodes $1, \rho, \ldots, \rho^d$ to the origin, or equivalently an exact $q$-Richardson filter. The same data determine every $C_m$ through a rational Vandermonde system. An accompanying Python program performs all calculations with exact rational arithmetic and exhaustively verifies the identities at all 257 reduced dyadic points of depth at most seven.

# Contents

# 1 Introduction

Finite truncations of the Fourier product of Rvachev's up-function are especially concrete. Each sinc factor is the Fourier transform of a centered uniform box, so the inverse transform of the first finitely many factors is a compactly supported piecewise-polynomial spline. The degree grows with the number of factors, the knots form a dyadic grid, and the signs of the top distributional derivative form the Thue–Morse sequence. The finite splines converge rapidly to the $C^\infty$ function up.

The natural expectation is that a finite spline value admits an asymptotic expansion in even powers of the omitted scale. The key observation of this article is that at a fixed dyadic rational this expansion becomes *exact and finite*. The reason is geometric rather than merely asymptotic. Once the finite-product grid is fine enough, the chosen dyadic point is the exact midpoint of one spline cell. The entire omitted tail has support equal to that cell's half-width. Consequently the convolution with the omitted tail sees one polynomial and nothing else. Deconvolution by the reciprocal sinc product is then an exact finite differential calculation on a finite-dimensional polynomial space. Finally, the dyadic derivative tiling of up forces every derivative above the dyadic depth to vanish at the point.

This produces a particularly simple answer to the extraction problem. If $x = a/2^s$ is in lowest dyadic terms and $d = \lfloor s/2 \rfloor$, then

$$Q_n(x) = P_x(4^{-n}) \qquad (n \geq s)$$

for one polynomial $P_x$ of degree exactly $d$ when $-1 < x < 1$ and $s \geq 2$. The desired value is its constant term,

$$\mathrm{up}(x) = P_x(0).$$

The samples $Q_{n_0}, \ldots, Q_{n_0+d}$ are values of $P_x$ on a geometric grid. The $q$-binomial theorem evaluates the corresponding Lagrange formula in closed form.

## 1.1 What is inherited and what is proved here

The `ProveIt` documentation develops the exact finite-spline formula, the self-similar head–tail factorization, the dyadic derivative-cell theorem, the reciprocal-product coefficients, an all-orders asymptotic expansion, and signed geometric Richardson acceleration. The present article combines those ingredients in a different way and proves the following strengthening:

> **Exact dyadic strengthening.** At a fixed dyadic point, the all-orders prefix expansion terminates exactly once the prefix reaches the dyadic depth. Therefore the general $q$-Richardson rule, which normally cancels only the first several asymptotic terms, returns the exact rational value after finitely many samples.

The proof below is analytic and uses only the standard infinite-product construction; the needed basic properties are proved in theorem 3.1. The computations supplied with the article are independent checks, not substitutes for proof. No claim is made here that the combined pointwise theorem has already been formalized in Lean.

## 1.2 Indexing convention

A one-step indexing shift is a common source of mistakes, so two notations will be kept throughout.

**Definition 1.1** (Factor-count and user indexing). For $N \geq 1$, define the $N$-factor prefix

$$\Phi_N(t) := \prod_{k=0}^{N-1} \operatorname{sinc}\left(\frac{\pi t}{2^k}\right), \qquad p_N := \mathcal{F}^{-1}\Phi_N. \tag{1.1}$$

The notation in the question includes factors $k = 0, \dots, n$, so

$$Q_n(x) = \mathrm{up}_n(x) = p_{n+1}(x). \tag{1.2}$$

The clean geometric statements use $p_N$; the final extraction formulas are stated for $Q_n$.

# 2 Fourier normalization and the finite box splines

## 2.1 Fourier convention and box factors

We use

$$\widehat{f}(t) = \int_{\mathbb{R}} f(x) e^{-2\pi i t x}\, \mathrm{d}x, \qquad f(x) = \int_{\mathbb{R}} \widehat{f}(t) e^{2\pi i t x}\, \mathrm{d}t \tag{2.1}$$

whenever the second formula is classically justified. Put

$$b_k(x) := 2^k \mathbf{1}_{[-2^{-k-1}, 2^{-k-1}]}(x), \qquad k \geq 0. \tag{2.2}$$

Each $b_k$ is a probability density and a direct integration gives

$$\widehat{b}_k(t) = \frac{\sin(\pi t/2^k)}{\pi t/2^k} = \operatorname{sinc}\left(\frac{\pi t}{2^k}\right). \tag{2.3}$$

Therefore

$$p_N = b_0 * b_1 * \cdots * b_{N-1}. \tag{2.4}$$

In probabilistic language, $p_N$ is the density of

$$S_N = \sum_{k=0}^{N-1} 2^{-k-1} U_k, \tag{2.5}$$

where the $U_k$ are independent and uniform on $[-1, 1]$. Its support is

$$\operatorname{supp} p_N = [-A_N, A_N], \qquad A_N := \sum_{k=0}^{N-1} 2^{-k-1} = 1 - 2^{-N}. \tag{2.6}$$

The infinite sum has density up and Fourier transform $\Phi$.

## 2.2 Exact truncated-power representation

Write

$$\tau_k := (-1)^{s_2(k)}, \tag{2.7}$$

where $s_2(k)$ is the number of 1's in the binary expansion of $k$. The following formula makes the spline structure and exact rational arithmetic explicit.

**Theorem 2.1** (Finite spline with Thue–Morse knots). *For every $N \geq 1$,*

$$\boxed{p_N(x) = \frac{2^{N(N-1)/2}}{(N-1)!} \sum_{k=0}^{2^N-1} \tau_k \left(x + A_N - \frac{k}{2^{N-1}}\right)_+^{N-1}.} \tag{2.8}$$

*The knots are*

$$t_{N,k} = -A_N + \frac{k}{2^{N-1}}, \qquad 0 \leq k < 2^N, \tag{2.9}$$

*and consecutive knots are separated by $2^{1-N}$.*

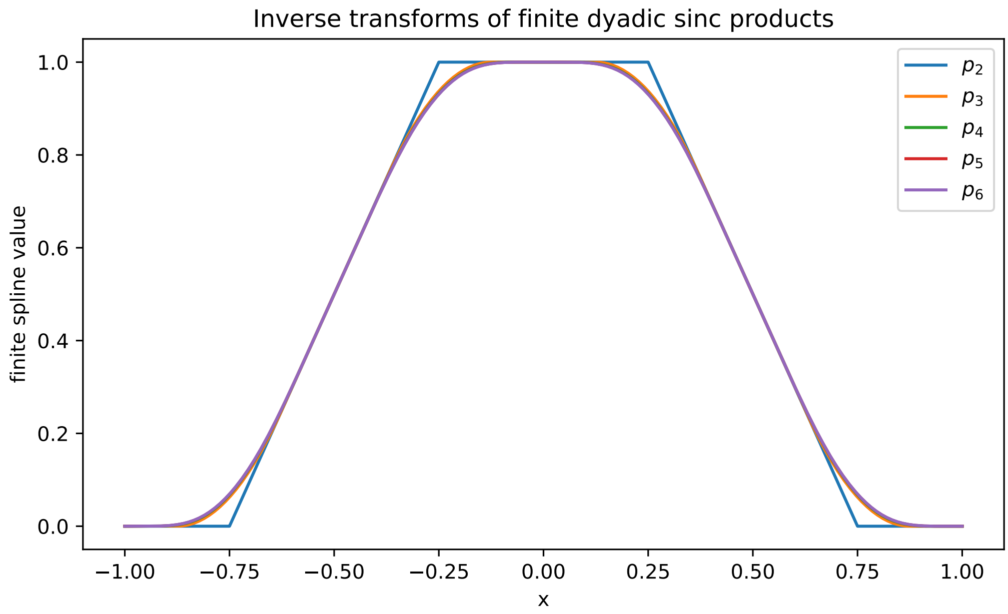


Figure 1: The first several finite inverse transforms $p_N$. Each new sinc factor convolves the preceding spline with a narrower centered box.

*Proof.* In the sense of distributions,

$$Db_k = 2^k \left(\delta_{-2^{-k-1}} - \delta_{2^{-k-1}}\right).$$

Differentiating the convolution (2.4) once in every factor gives

$$D^N p_N = (Db_0) * (Db_1) * \cdots * (Db_{N-1}).$$

Choose a binary vector $(\beta_0, \dots, \beta_{N-1}) \in \{0,1\}^N$, where $\beta_j = 0$ selects the negative endpoint and $\beta_j = 1$ the positive endpoint. Starting from the all-negative position $-A_N$, changing the $j$-th choice adds $2^{-j}$. Hence the resulting atom is at

$$-A_N + \sum_{j=0}^{N-1} \beta_j 2^{-j} = -A_N + \frac{k}{2^{N-1}}, \qquad k = \sum_{j=0}^{N-1} \beta_j 2^{N-1-j}.$$

The sign is $(-1)^{\sum \beta_j} = \tau_k$ because reversing binary digits does not change their sum. The common magnitude is

$$\prod_{j=0}^{N-1} 2^j = 2^{N(N-1)/2}.$$

Thus

$$D^N p_N = 2^{N(N-1)/2} \sum_{k=0}^{2^N-1} \tau_k \delta_{t_{N,k}}.$$

Since

$$D^N \frac{(x-t)_+^{N-1}}{(N-1)!} = \delta_t,$$

integrating $N$ times from $-\infty$ and using compact support yields (2.8). $\square$

**Corollary 2.2** (Scaled cell formula). *Put*

$$y = 2^{N-1}(x + A_N). \tag{2.10}$$

*Inside the support and away from the harmless $N = 1$ jump convention,*

$$p_N(x) = \frac{2^{-(N-1)(N-2)/2}}{(N-1)!} \sum_{0\le k\le \lfloor y\rfloor} \tau_k (y-k)^{N-1}. \tag{2.11}$$

*In particular, $p_N(x) \in \mathbb{Q}$ whenever $x \in \mathbb{Q}$.*

*Proof.* Factor $2^{-(N-1)}$ from every truncated power in (2.8). Rationality is immediate. □

### 2.3 Why dyadic points become cell midpoints

**Definition 2.3** (Dyadic depth). For a dyadic rational $x$, define

$$\mathrm{depth}_2(x) := \min\{s \ge 0 : 2^s x \in \mathbb{Z}\}. \tag{2.12}$$

Thus a nonintegral dyadic has a unique reduced form $x = a/2^s$ with $a$ odd and $s = \mathrm{depth}_2(x)$.

**Lemma 2.4** (Exact midpoint alignment). *Let $-1 < x < 1$ be dyadic of depth $s$. For every $N \ge s+1$, the point $x$ is the midpoint of a unique polynomial cell of $p_N$, and that cell is precisely*

$$[x - 2^{-N},\, x + 2^{-N}]. \tag{2.13}$$

*Proof.* By (2.9), the cell width is $2^{1-N}$ and its half-width is $2^{-N}$. Its midpoints are

$$\frac{t_{N,k} + t_{N,k+1}}{2} = -1 + \frac{k+1}{2^{N-1}}.$$

Since $N - 1 \ge s$, the number

$$J = 2^{N-1}(1 + x)$$

is an integer. Because $-1 < x < 1$, one has $1 \le J \le 2^N - 1$. Taking $k = J - 1$ gives the required midpoint. The claimed endpoints follow from the half-width. □

This exact alignment is the first reason the eventual law is finite. It is not true at a generic irrational point: as $N$ changes, the omitted tail generally straddles a knot instead of remaining inside one polynomial piece.

# 3 Head–tail self-similarity

### 3.1 The omitted tail is a scaled copy of the whole function

Set

$$\delta_N := 2^{-N}. \tag{3.1}$$

Splitting the infinite product after the first $N$ factors gives

$$\Phi(t) = \Phi_N(t)\Phi(\delta_N t). \tag{3.2}$$

Define the scaled density

$$\mathrm{up}_\delta(y) := \delta^{-1}\mathrm{up}(y/\delta). \tag{3.3}$$

Its Fourier transform is $\Phi(\delta t)$, and it is supported on $[-\delta, \delta]$. Fourier inversion of (3.2) yields

$$\boxed{\mathrm{up} = p_N * \mathrm{up}_{\delta_N}.} \tag{3.4}$$

Probabilistically, the omitted random sum is $\delta_N$ times an independent copy of the full random sum.

**Proposition 3.1** (Basic analytic properties and strict positivity). *The function* up *is an even, nonnegative $C^\infty$ probability density with support* $[-1,1]$. *Moreover,*

$$\mathrm{up}(0)=1,\qquad \mathrm{up}(\pm1)=0,\qquad \mathrm{up}(x)>0\quad(-1<x<1), \tag{3.5}$$

*and it satisfies*

$$\mathrm{up}'(x)=2\big(\mathrm{up}(2x+1)-\mathrm{up}(2x-1)\big). \tag{3.6}$$

*Proof.* The random series in (2.5), continued to infinity, converges almost surely and takes values in $[-1,1]$. Its law is even and is the convolution of the first box $b_0$ with the law of the remaining tail, hence it has a nonnegative density of total mass one and Fourier transform $\Phi$. For every fixed $K$, the first $K$ sinc factors give $|\Phi(t)|\le C_K(1+|t|)^{-K}$, while all remaining factors have modulus at most one. Fourier inversion therefore gives derivatives of every order, so the density is $C^\infty$.

The tail after $b_0$ is supported on $[-1/2,1/2]$, where $b_0$ is identically one; consequently $\mathrm{up}(0)=1$. Continuity and compact support give $\mathrm{up}(\pm1)=0$. Each finite convolution $p_N$ is strictly positive in the interior of its support: if two nonnegative densities are positive on the interiors of intervals, then their convolution is positive on the interior of the sum interval because the overlap of the two positivity intervals has positive length. Now fix $|x|<1$ and choose $N$ so large that $2\delta_N<1-|x|$. For every $|y|\le\delta_N$,

$$|x-y|<1-\delta_N=A_N,$$

so $p_N(x-y)>0$. The head–tail convolution (3.4) then implies $\mathrm{up}(x)>0$.

Finally, splitting off the first box gives $\mathrm{up}=b_0*\mathrm{up}_{1/2}$ and hence

$$\mathrm{up}(x)=\int_{x-1/2}^{x+1/2}2\mathrm{up}(2y)\,\mathrm{d}y=\int_{2x-1}^{2x+1}\mathrm{up}(z)\,\mathrm{d}z.$$

Differentiation proves (3.6). ☐

### 3.2 The reciprocal sinc coefficients

Near the origin define rational coefficients $a_m$ by

$$\frac{1}{\Phi(z)}=\exp\left(\sum_{j=1}^{\infty}\alpha_j(2\pi z)^{2j}\right)=\sum_{m=0}^{\infty}a_m(2\pi z)^{2m}. \tag{3.7}$$

The logarithmic expansion of $w/\sin w$ gives

$$\alpha_j=\frac{|B_{2j}|}{2j(2j)!\,(1-2^{-2j})}, \tag{3.8}$$

where $B_{2j}$ are Bernoulli numbers. Consequently

$$a_0=1,\qquad ma_m=\sum_{j=1}^{m}j\alpha_ja_{m-j}. \tag{3.9}$$

Equivalently, with $Y_m$ denoting the complete exponential Bell polynomial,

$$a_m=\frac{1}{m!}Y_m(1!\alpha_1,2!\alpha_2,\dots,m!\alpha_m). \tag{3.10}$$

The first values are

$$\begin{aligned} a_0&=1, & a_1&=\frac{1}{18}, & a_2&=\frac{31}{16200},\\ a_3&=\frac{2347}{42865200}, & a_4&=\frac{1904369}{1311675120000}, & a_5&=\frac{3309760579}{88561680752160000}. \end{aligned} \tag{3.11}$$

Every $\alpha_j$ is positive, so (3.9) also shows $a_m>0$.

*Derivation of* (3.8). The standard Bernoulli expansion is

$$\log \frac{w}{\sin w} = \sum_{j=1}^{\infty} \frac{2^{2j-1}\,|B_{2j}|}{j(2j)!} w^{2j}.$$

Apply this to $w = \pi z/2^k$ and sum over $k \geq 0$. The geometric sum contributes $(1-2^{-2j})^{-1}$. Replacing $(\pi z)^{2j}$ by $2^{-2j}(2\pi z)^{2j}$ gives (3.8). Differentiating the formal identity in (3.7) with respect to $X = (2\pi z)^2$ and comparing coefficients yields (3.9); the Bell-polynomial form is the exponential formula. □

# 4 Exact polynomial deconvolution on one cell

The usual Fourier argument gives an asymptotic differential expansion of $p_N$ in powers of $\delta_N^2$. At an aligned dyadic point we can do better because we are deconvolving a genuine polynomial on the whole support of the tail.

## 4.1 A finite-dimensional operator identity

Let

$$M(z) := \int_{-1}^{1} \mathrm{up}(y) e^{zy}\,\mathrm{d}y. \tag{4.1}$$

By the Fourier convention,

$$M(z) = \Phi\left(\frac{iz}{2\pi}\right), \qquad \frac{1}{M(z)} = \sum_{m=0}^{\infty} (-1)^m a_m z^{2m}. \tag{4.2}$$

**Lemma 4.1** (Exact inverse on polynomials)**.** *Let $P$ be a polynomial of degree at most $D$, let $\delta > 0$, and define*

$$(T_\delta P)(x) := \int_{-1}^{1} \mathrm{up}(y) P(x - \delta y)\,\mathrm{d}y. \tag{4.3}$$

*Then*

$$\boxed{P(x) = \sum_{m=0}^{\lfloor D/2 \rfloor} (-1)^m a_m \delta^{2m} (T_\delta P)^{(2m)}(x).} \tag{4.4}$$

*The identity is algebraic and exact; no limiting argument and no analytic convergence of the infinite reciprocal series are needed.*

*Proof.* Let $D_x$ denote differentiation with respect to $x$. Taylor's formula is finite on polynomials, so

$$P(x - \delta y) = e^{-\delta y D_x} P(x)$$

can be interpreted as a finite sum. Integrating gives

$$T_\delta = M(-\delta D_x) = M(\delta D_x),$$

because up and hence $M$ are even. On the vector space of polynomials of degree at most $D$, the operator $D_x$ is nilpotent. Therefore the formal identity

$$M(z)^{-1} = \sum_{m \geq 0} (-1)^m a_m z^{2m}$$

may be substituted at $z = \delta D_x$ and truncates automatically after $m = \lfloor D/2 \rfloor$. This gives

$$T_\delta^{-1} = \sum_{m=0}^{\lfloor D/2 \rfloor} (-1)^m a_m \delta^{2m} D_x^{2m},$$

which is (4.4). □

### 4.2 Locality converts the inverse into an identity for $p_N$

**Lemma 4.2** (The tail sees only one polynomial piece)**.** *Suppose $x$ is the midpoint of a cell of $p_N$, and let $P$ be the polynomial of degree at most $N-1$ that agrees with $p_N$ on that cell. Then, for every $r \geq 0$,*

$$(P * \mathrm{up}_{\delta_N})^{(r)}(x) = \mathrm{up}^{(r)}(x). \tag{4.5}$$

*Proof.* The cell is $[x-\delta_N, x+\delta_N]$, while $\operatorname{supp} \mathrm{up}_{\delta_N} = [-\delta_N, \delta_N]$. Differentiating the smooth compactly supported kernel in the convolution gives

$$\mathrm{up}^{(r)}(x) = \int_{\mathbb{R}} p_N(z) \mathrm{up}_{\delta_N}^{(r)}(x-z)\,\mathrm{d}z.$$

Only $z \in [x-\delta_N, x+\delta_N]$ contributes, and there $p_N(z) = P(z)$. Replacing $p_N$ by $P$ therefore leaves the integral unchanged, proving (4.5). □

**Theorem 4.3** (Exact local differential formula)**.** *If $x$ is the midpoint of a cell of $p_N$, then*

$$\boxed{p_N(x) = \sum_{m=0}^{\lfloor (N-1)/2 \rfloor} (-1)^m a_m 4^{-mN} \mathrm{up}^{(2m)}(x).} \tag{4.6}$$

*More generally, for every $r \leq N-1$,*

$$p_N^{(r)}(x) = \sum_{m=0}^{\lfloor (N-1-r)/2 \rfloor} (-1)^m a_m 4^{-mN} \mathrm{up}^{(r+2m)}(x). \tag{4.7}$$

*Proof.* With $P$ as in the preceding lemma, scaling in (4.3) shows $T_{\delta_N} P = P * \mathrm{up}_{\delta_N}$. Apply (4.4) with $D = N-1$, evaluate at $x$, and use (4.5). Since $\delta_N^{2m} = 4^{-mN}$, this gives (4.6). Differentiate the polynomial operator identity $r$ times, or apply it to $P^{(r)}$, to obtain (4.7). □

*Remark* 4.4. For general $x$, the analogous formula is an all-orders asymptotic expansion. At a cell midpoint it is exact because the nonlocal Fourier multiplier has been reduced to a finite-dimensional polynomial operator. This is the decisive upgrade from asymptotic cancellation to exact extraction.

## 5 Dyadic derivative tiling and termination

### 5.1 The exact derivative-cell formula

The Rvachev functional-differential equation is

$$\mathrm{up}'(x) = 2\big(\mathrm{up}(2x+1) - \mathrm{up}(2x-1)\big). \tag{5.1}$$

Iterating it produces a global Thue–Morse sum.

**Lemma 5.1** (Iterated derivative sum)**.** *For every $r \geq 0$ and $x \in \mathbb{R}$,*

$$\mathrm{up}^{(r)}(x) = 2^{\binom{r+1}{2}} \sum_{k=0}^{2^r-1} \tau_k \, \mathrm{up}(2^r(x+1) - (2k+1)). \tag{5.2}$$

*Proof.* For $r=0$ the sum has one term and is $\mathrm{up}(x)$. Assume the formula at order $r$. Differentiate; the chain rule contributes $2^r$, and then apply (5.1) to every summand. The positive branch has new index $2k$ and sign $\tau_{2k} = \tau_k$; the negative branch has new index $2k+1$ and sign $\tau_{2k+1} = -\tau_k$. The factor gains $2^{r+1}$, changing $2^{\binom{r+1}{2}}$ to $2^{\binom{r+2}{2}}$. This is the formula at order $r+1$. □

For $0 \le k < 2^r$ and $-1 \le y \le 1$, define the cell coordinate

$$c_{r,k}(y) := -1 + \frac{2k+1+y}{2^r}. \tag{5.3}$$

**Theorem 5.2** (Exact derivative tiling)**.** *For every $r \ge 0$, $0 \le k < 2^r$, and $-1 \le y \le 1$,*

$$\boxed{\mathrm{up}^{(r)}\big(c_{r,k}(y)\big) = 2^{\binom{r+1}{2}} \tau_k \mathrm{up}(y).} \tag{5.4}$$

*In particular,*

$$\mathrm{up}^{(r)}\left(-1 + \frac{2k+1}{2^r}\right) = 2^{\binom{r+1}{2}} \tau_k. \tag{5.5}$$

*Proof.* Insert $x = c_{r,k}(y)$ into (5.2). The argument in the $j$-th summand becomes

$$2^r(c_{r,k}(y)+1) - (2j+1) = 2(k-j) + y.$$

Because up is supported on $[-1,1]$, only $j = k$ can contribute. At $y = \pm 1$ a neighboring argument can also lie at an endpoint, but $\mathrm{up}(\pm 1) = 0$, so the same conclusion remains valid. This proves (5.4). Set $y = 0$ and use $\mathrm{up}(0) = 1$ for (5.5). □

## 5.2 The dyadic jet stops at the dyadic depth

**Theorem 5.3** (Dyadic derivative cutoff)**.** *Let $-1 < x < 1$ be dyadic of depth $s \ge 1$. Then*

$$\mathrm{up}^{(r)}(x) \ne 0 \qquad (0 \le r \le s), \tag{5.6}$$

$$\mathrm{up}^{(r)}(x) = 0 \qquad (r > s). \tag{5.7}$$

*At the top nonzero order,*

$$\mathrm{up}^{(s)}(x) = 2^{\binom{s+1}{2}} \tau_k, \qquad k = \frac{2^s(1+x)-1}{2}. \tag{5.8}$$

*Proof.* Write $x = a/2^s$ in lowest terms, so $a$ is odd. At $r = s$,

$$2^s(1+x) = 2^s + a$$

is odd. Hence it is $2k+1$ for an integer $k$, and $x = c_{s,k}(0)$. Equation (5.5) proves (5.8).

If $r > s$, then $2^r(1+x) = 2^{r-s}(2^s + a)$ is an even integer, say $2h$. Thus $x$ is a boundary of two order-$r$ derivative cells:

$$x = c_{r,h-1}(1) = c_{r,h}(-1).$$

Applying (5.4) and $\mathrm{up}(\pm 1) = 0$ gives (5.7).

Finally, if $r < s$, the number $2^r(1+x)$ is not an integer. It belongs to the interior of one order-$r$ cell, so $x = c_{r,k}(y)$ with $-1 < y < 1$. The function up is strictly positive on $(-1,1)$; hence (5.4) is nonzero. The case $r = 0$ is the same positivity statement. □

The function is $C^\infty$, yet its full derivative jet at a dyadic point is finite: it ends exactly at the denominator depth. This exceptional flatness is the second reason the prefix expansion terminates.

## 6 The eventual geometric law

We now combine midpoint alignment, exact local deconvolution, and the derivative cutoff.

**Theorem 6.1** (Exact eventual polynomialization)**.** *Let $x \in [-1, 1]$ be dyadic, let*

$$s = \operatorname{depth}_2(x), \qquad d = \lfloor s/2 \rfloor, \qquad \rho = \frac{1}{4}, \tag{6.1}$$

*and define $Q_n(x) = p_{n+1}(x)$. There exist constants $C_1(x), \dots, C_d(x)$ such that*

$$\boxed{Q_n(x) = \operatorname{up}(x) + \sum_{m=1}^{d} C_m(x)\rho^{mn} \qquad (n \geq s).} \tag{6.2}$$

*They are explicitly*

$$\boxed{C_m(x) = (-1)^m a_m \rho^m \operatorname{up}^{(2m)}(x) = (-1)^m a_m 4^{-m} \operatorname{up}^{(2m)}(x).} \tag{6.3}$$

*For $-1 < x < 1$ and $s \geq 2$, all $d$ correction coefficients are nonzero, so the polynomial*

$$P_x(z) := \operatorname{up}(x) + \sum_{m=1}^{d} C_m(x) z^m \tag{6.4}$$

*has degree exactly $d$ and*

$$Q_n(x) = P_x(4^{-n}) \qquad (n \geq s). \tag{6.5}$$

*Proof.* First suppose $-1 < x < 1$ and $s \geq 1$. For $n \geq s$, put $N = n + 1$. Then $N \geq s + 1$, so $x$ is a cell midpoint of $p_N$ by the midpoint-alignment lemma. The exact local formula (4.6) gives

$$Q_n(x) = p_{n+1}(x) = \sum_{m=0}^{\lfloor n/2 \rfloor} (-1)^m a_m 4^{-m(n+1)} \operatorname{up}^{(2m)}(x).$$

Because $n \geq s$, every even order $2m \leq s$ occurs in the displayed sum. Every term with $2m > s$ vanishes by (5.7). Hence the surviving indices are exactly $0 \leq m \leq \lfloor s/2 \rfloor = d$, and

$$Q_n(x) = \operatorname{up}(x) + \sum_{m=1}^{d} \Big((-1)^m a_m 4^{-m} \operatorname{up}^{(2m)}(x)\Big) 4^{-mn}.$$

This proves (6.2) and (6.3).

If $s \geq 2$, then $2m \leq s$ for $1 \leq m \leq d$. The derivative cutoff theorem says these even derivatives are nonzero, and $a_m > 0$, so every $C_m(x)$ is nonzero.

It remains to cover depth $s = 0$. The possible points in $[-1, 1]$ are $-1, 0, 1$. Every finite spline vanishes at $\pm 1$ because its support is $[-A_N, A_N]$ with $A_N < 1$, while $\operatorname{up}(\pm 1) = 0$. At the origin, write $p_N = b_0 * (b_1 * \cdots * b_{N-1})$. The second factor is a probability density supported in $[-1/2 + 2^{-N}, 1/2 - 2^{-N}]$, where the translate seen by $b_0$ has constant value 1; hence $p_N(0) = 1 = \operatorname{up}(0)$. Thus the formula holds with $d = 0$. □

**Corollary 6.2** (Number and ratios of the geometric corrections)**.** *At a reduced dyadic point of depth $s$, at most the first $s$ user-indexed values $Q_0, \dots, Q_{s-1}$ are exceptional. Thereafter there are $d = \lfloor s/2 \rfloor$ correction modes. The $m$-th mode has common ratio*

$$\frac{C_m 4^{-m(n+1)}}{C_m 4^{-mn}} = 4^{-m}. \tag{6.6}$$

*Thus each individual correction decreases geometrically in absolute value, although cancellation between modes need not make the total error monotone.*

### 6.1 A cell formula for the correction coefficients

For $1 \le m \le d$, set $r = 2m$ and define

$$k_m = \left\lfloor 2^{2m-1}(1+x) \right\rfloor, \qquad y_m = 2^{2m}(1+x) - (2k_m + 1). \tag{6.7}$$

Then $-1 < y_m < 1$; for $2m < s$ it lies strictly in $(-1, 1)$, and for $2m = s$ it is $0$. From the derivative tiling,

$$\mathrm{up}^{(2m)}(x) = 2^{m(2m+1)} \tau_{k_m} \mathrm{up}(y_m). \tag{6.8}$$

Substitution into (6.3) gives

$$\boxed{C_m(x) = (-1)^m a_m 2^{2m^2-m} \tau_{k_m} \mathrm{up}(y_m).} \tag{6.9}$$

This expresses every geometric coefficient in terms of a lower-scale dyadic value of the same function. Rationality will also follow directly from the extraction theorem below, so (6.9) then shows that all $C_m(x)$ are rational.

### 6.2 Derivative versions

The same argument applies to derivatives of the finite splines.

**Corollary 6.3** (Eventual law for derivative approximants)**.** *Let $0 \le r \le s$, and evaluate the cellwise derivative of $p_{n+1}$ at $x$. For every $n \ge s$,*

$$p_{n+1}^{(r)}(x) = \mathrm{up}^{(r)}(x) + \sum_{m=1}^{\lfloor (s-r)/2 \rfloor} (-1)^m a_m 4^{-m} \mathrm{up}^{(r+2m)}(x)\, 4^{-mn}. \tag{6.10}$$

*Hence the same $q$-extraction formula recovers $\mathrm{up}^{(r)}(x)$, with $d$ replaced by $\lfloor (s-r)/2 \rfloor$.*

*Proof.* Use (4.7) and terminate it with (5.7). □

## 7 Exact extraction by a $q$-Pochhammer filter

### 7.1 Shift-operator proof

Let $E$ be the forward shift on sequences:

$$(Ef)_n = f_{n+1}. \tag{7.1}$$

The constant sequence is an eigenvector of $E$ with eigenvalue $1$, and the mode $\rho^{mn}$ is an eigenvector with eigenvalue $\rho^m$. Therefore the polynomial

$$\mathcal{P}_d(E) := \prod_{m=1}^{d} (E - \rho^m) \tag{7.2}$$

annihilates every correction mode in (6.2). On the constant term it acts by

$$\mathcal{P}_d(1) = \prod_{m=1}^{d} (1 - \rho^m) = (\rho; \rho)_d. \tag{7.3}$$

Here

$$(a; q)_d := \prod_{r=0}^{d-1} (1 - aq^r), \qquad (q; q)_d = \prod_{r=1}^{d} (1 - q^r) \tag{7.4}$$

is the finite $q$-Pochhammer symbol.

The finite $q$-binomial theorem gives

$$\prod_{m=1}^{d}(z-\rho^m)=\sum_{j=0}^{d}(-1)^{d-j}\rho^{\binom{d-j+1}{2}}\begin{bmatrix}d\\j\end{bmatrix}_\rho z^j, \tag{7.5}$$

where

$$\begin{bmatrix}d\\j\end{bmatrix}_q := \frac{(q;q)_d}{(q;q)_j(q;q)_{d-j}} \tag{7.6}$$

is the Gaussian binomial coefficient.

**Theorem 7.1** (Single-formula exact extractor). *Under the hypotheses of theorem 6.1, let $n_0 \geq s$. Then*

$$\boxed{\mathrm{up}(x)=\frac{1}{(\rho;\rho)_d}\sum_{j=0}^{d}(-1)^{d-j}\rho^{\binom{d-j+1}{2}}\begin{bmatrix}d\\j\end{bmatrix}_\rho Q_{n_0+j}(x), \qquad \rho=\frac{1}{4}.} \tag{7.7}$$

*Equivalently,*

$$\boxed{\mathrm{up}(x)=\sum_{j=0}^{d}\frac{(-1)^{d-j}\rho^{\binom{d-j+1}{2}}}{(\rho;\rho)_j(\rho;\rho)_{d-j}}Q_{n_0+j}(x).} \tag{7.8}$$

*Proof.* Apply (7.2) to (6.2) at index $n_0$. Every mode with $1 \leq m \leq d$ is killed by its factor $E-\rho^m$, while the constant survives with multiplier $(\rho;\rho)_d$. Hence

$$\mathcal{P}_d(E)Q_{n_0}=(\rho;\rho)_d\mathrm{up}(x).$$

Expand $\mathcal{P}_d$ by (7.5) and divide by $(\rho;\rho)_d$ to get (7.7). Substitute (7.6) for (7.8). □

**Corollary 7.2** (One completely explicit finite formula). *Let $N_j=n_0+j+1$ and $A_{N_j}=1-2^{-N_j}$. Substitution of the exact spline formula into (7.8) gives*

$$\boxed{\begin{aligned}\mathrm{up}(x)=\sum_{j=0}^{d}&\frac{(-1)^{d-j}\rho^{\binom{d-j+1}{2}}}{(\rho;\rho)_j(\rho;\rho)_{d-j}}\frac{2^{N_j(N_j-1)/2}}{(N_j-1)!}\\ &\times\sum_{k=0}^{2^{N_j}-1}\tau_k\left(x+1-2^{-N_j}-\frac{k}{2^{N_j-1}}\right)_+^{N_j-1}, \qquad \rho=\frac{1}{4}.\end{aligned}} \tag{7.9}$$

*Every sum and product in (7.9) is finite. Taking $n_0=s$ gives the smallest guaranteed formula, involving exactly $d+1$ finite splines.*

*Proof.* Use $Q_{n_0+j}=p_{N_j}$ in (7.8) and substitute (2.8). No passage to a limit remains. □

## 7.2 Integer-coefficient base-4 form

The inversion identity

$$\begin{bmatrix}d\\j\end{bmatrix}_{q^{-1}}=q^{-j(d-j)}\begin{bmatrix}d\\j\end{bmatrix}_q \tag{7.10}$$

and

$$(1/4;1/4)_d=4^{-d(d+1)/2}\prod_{r=1}^{d}(4^r-1) \tag{7.11}$$

convert (7.7) to an especially useful rational formula.

**Corollary 7.3** (Common-denominator integer formula). *For every* $n_0 \geq s$*,*

$$\boxed{\operatorname{up}(x) = \frac{\displaystyle\sum_{j=0}^{d} (-1)^{d-j} 4^{\binom{j+1}{2}} \begin{bmatrix} d \\ j \end{bmatrix}_4 Q_{n_0+j}(x)}{\displaystyle\prod_{r=1}^{d} (4^r - 1)}.} \tag{7.12}$$

*All coefficients in the numerator are integers and the common denominator is odd.*

*Proof.* Insert (7.10) with $q = 4$ and (7.11) into (7.7). The total power of 4 multiplying the $j$-th term is

$$\binom{d+1}{2} - \binom{d-j+1}{2} - j(d-j) = \binom{j+1}{2}.$$

Gaussian binomial coefficients at the integer base 4 are integers. □

**Corollary 7.4** (Rationality at dyadic points). *For every dyadic* $x \in [-1, 1]$*, one has* $\operatorname{up}(x) \in \mathbb{Q}$*.*

*Proof.* Every finite value $Q_{n_0+j}(x)$ is rational by the truncated-power formula, and every extraction weight is rational. Equation (7.8) finishes the proof. □

This gives a direct proof of classical dyadic rationality from finite Fourier-product splines.

### 7.3 First weight vectors

Writing

$$\operatorname{up}(x) = \sum_{j=0}^{d} W_{d,j} Q_{n_0+j}(x), \tag{7.13}$$

the first vectors are

$$d = 0: \quad (1), \tag{7.14}$$

$$d = 1: \quad \left(-\frac{1}{3}, \frac{4}{3}\right), \tag{7.15}$$

$$d = 2: \quad \left(\frac{1}{45}, -\frac{4}{9}, \frac{64}{45}\right), \tag{7.16}$$

$$d = 3: \quad \left(-\frac{1}{2835}, \frac{4}{135}, -\frac{64}{135}, \frac{4096}{2835}\right), \tag{7.17}$$

$$d = 4: \quad \left(\frac{1}{722925}, -\frac{4}{8505}, \frac{64}{2025}, -\frac{4096}{8505}, \frac{1048576}{722925}\right). \tag{7.18}$$

The weights depend only on the denominator depth through $d$, not on the numerator of $x$ and not on the chosen valid starting index $n_0$.

### 7.4 Stability of the signed formula

Alternating extrapolation weights can sometimes be numerically dangerous. Here the total variation is uniformly small.

**Proposition 7.5** (Uniform weight bound). *For* $0 < \rho < 1$,

$$\sum_{j=0}^{d} |W_{d,j}| = \frac{(-\rho;\rho)_d}{(\rho;\rho)_d}. \tag{7.19}$$

*At* $\rho = 1/4$,

$$\sum_{j=0}^{d} |W_{d,j}| < \frac{(-1/4;1/4)_\infty}{(1/4;1/4)_\infty} = 1.969260353668269\ldots. \tag{7.20}$$

*Consequently, if each input value has absolute error at most* $\eta$, *the extracted value has error less than* $1.97\eta$.

*Proof.* Use (7.7), discard the alternating signs, and replace $j$ by $d-r$:

$$\sum_{j=0}^{d} |W_{d,j}| = \frac{1}{(\rho;\rho)_d} \sum_{r=0}^{d} \rho^{r(r+1)/2} \begin{bmatrix} d \\ r \end{bmatrix}_\rho.$$

The finite $q$-binomial theorem with positive signs says that the numerator is

$$\prod_{m=1}^{d} (1+\rho^m) = (-\rho;\rho)_d.$$

The finite products increase to the displayed infinite-product ratio. The error statement is the triangle inequality. ☐

# 8 The Vandermonde system and recovery of every coefficient

The extraction formula gives the constant without solving a system explicitly. The full geometric law is recovered from exactly the same data.

## 8.1 Polynomial interpolation on geometric nodes

Fix a valid starting index $n_0 \geq s$ and define

$$A_m := C_m(x)\rho^{mn_0}, \qquad R(z) := \mathrm{up}(x) + \sum_{m=1}^{d} A_m z^m. \tag{8.1}$$

Then

$$R(\rho^j) = Q_{n_0+j}(x), \qquad 0 \leq j \leq d. \tag{8.2}$$

Therefore the unknown vector $(\mathrm{up}(x), A_1, \ldots, A_d)^T$ is the solution of

$$\begin{pmatrix} 1 & 1 & 1 & \cdots & 1 \\ 1 & \rho & \rho^2 & \cdots & \rho^d \\ 1 & \rho^2 & \rho^4 & \cdots & \rho^{2d} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & \rho^d & \rho^{2d} & \cdots & \rho^{d^2} \end{pmatrix} \begin{pmatrix} \mathrm{up}(x) \\ A_1 \\ A_2 \\ \vdots \\ A_d \end{pmatrix} = \begin{pmatrix} Q_{n_0}(x) \\ Q_{n_0+1}(x) \\ Q_{n_0+2}(x) \\ \vdots \\ Q_{n_0+d}(x) \end{pmatrix}. \tag{8.3}$$

The determinant is

$$\prod_{0\leq a<b\leq d} (\rho^b - \rho^a) \neq 0, \tag{8.4}$$

so the solution is unique over $\mathbb{Q}$.

The Lagrange form is

$$R(z) = \sum_{j=0}^{d} Q_{n_0+j}(x)\ell_j(z), \qquad \ell_j(z) = \prod_{\substack{0\le r\le d\\ r\ne j}} \frac{z-\rho^r}{\rho^j-\rho^r}. \tag{8.5}$$

Thus every coefficient is available in one expression:

$$\boxed{C_m(x) = \rho^{-mn_0}\sum_{j=0}^{d} Q_{n_0+j}(x)[z^m]\ell_j(z), \qquad 0\le m\le d,} \tag{8.6}$$

where $C_0(x) = \mathrm{up}(x)$. A convenient $q$-Pochhammer normalization of the Lagrange denominator is

$$\prod_{\substack{0\le r\le d\\ r\ne j}} (\rho^j-\rho^r) = (-1)^j\rho^{j(j-1)/2+j(d-j)}(\rho;\rho)_j(\rho;\rho)_{d-j}. \tag{8.7}$$

Setting $z = 0$ in (8.5) reduces exactly to (7.8).

### 8.2 Minimality

**Proposition 8.1** (Why $d+1$ samples are generically necessary). *For an interior dyadic point of depth $s\ge 2$, the $d+1$ sequences*

$$1, \rho^n, \rho^{2n}, \dots, \rho^{dn} \tag{8.8}$$

*all occur with nonzero coefficients in (6.2) and are linearly independent. Consequently, without additional derivative information, $d+1$ scalar samples are the minimal generic amount of data needed to identify all coefficients, including* $\mathrm{up}(x)$.

*Proof.* Nonvanishing was proved in theorem 6.1. Linear independence follows by evaluating a putative relation at $d+1$ consecutive integers; the resulting coefficient matrix is the nonsingular Vandermonde matrix (8.3). Fewer than $d+1$ equations cannot uniquely determine $d+1$ unconstrained coefficients. □

## 9 Annihilating recurrences and onset diagnostics

Because the eventual sequence is a finite sum of exponentials, it satisfies a constant-coefficient recurrence.

**Theorem 9.1** (Exact eventual recurrence). *For every $n\ge s$,*

$$\boxed{\prod_{m=0}^{d}(E-\rho^m)Q_n(x) = 0.} \tag{9.1}$$

*Equivalently,*

$$\sum_{j=0}^{d+1}(-1)^{d+1-j}\rho^{\binom{d+1-j}{2}}\begin{bmatrix} d+1\\ j\end{bmatrix}_\rho Q_{n+j}(x) = 0. \tag{9.2}$$

*Proof.* The factor $E-1$ kills the constant term, and $E-\rho^m$ kills the $m$-th geometric mode. The expanded form is the finite $q$-binomial theorem. □

This recurrence is useful in three ways.

1. It gives an exact check that a proposed starting index is late enough.
2. If a dyadic fraction has not first been reduced, the smallest recurrence order detected from exact data reveals the effective value $d = \lfloor s/2 \rfloor$.
3. In floating-point calculations, its residual measures both evaluation error and contamination by pre-asymptotic values.

The recurrence order is $d + 1$ because it includes the constant eigenvalue 1. The extraction filter in theorem 7.1 omits $E - 1$, preserving the constant while annihilating all error modes.

# 10 Worked examples

## 10.1 Depth zero and one

At $x = 0$, every $Q_n(0) = 1$, so no extrapolation is needed. At $x = \pm 1$, every $Q_n(x) = 0$. For $x = \pm 1/2$, the depth is $s = 1$ and $d = 0$, hence

$$Q_n(\pm 1/2) = \mathrm{up}(\pm 1/2) = \frac{1}{2} \qquad (n \geq 1). \tag{10.1}$$

The possibly convention-dependent endpoint value of the first box $Q_0$ is one of the allowed initial exceptions.

## 10.2 One correction mode: $x = 3/4$

Here $s = 2$, $d = 1$, and the exact finite values are

$$Q_2\left(\frac{3}{4}\right) = \frac{1}{16}, \qquad Q_3\left(\frac{3}{4}\right) = \frac{13}{192}. \tag{10.2}$$

The $d = 1$ extractor is

$$\mathrm{up}(x) = \frac{4Q_{n_0+1}(x) - Q_{n_0}(x)}{3}. \tag{10.3}$$

At $n_0 = 2$ it gives

$$\mathrm{up}\left(\frac{3}{4}\right) = \frac{4(13/192) - 1/16}{3} = \frac{5}{72}. \tag{10.4}$$

Moreover $3/4$ is the midpoint of the $k = 3$ second-derivative cell, so

$$\mathrm{up}''\left(\frac{3}{4}\right) = 2^{\binom{3}{2}} \tau_3 \mathrm{up}(0) = 8. \tag{10.5}$$

Since $a_1 = 1/18$,

$$C_1 = -\frac{1}{18}\frac{1}{4} \cdot 8 = -\frac{1}{9}. \tag{10.6}$$

Therefore the complete tail is

$$\boxed{Q_n\left(\frac{3}{4}\right) = \frac{5}{72} - \frac{1}{9}4^{-n} \qquad (n \geq 2).} \tag{10.7}$$

The threshold is sharp here: the displayed law does not hold at $n = 1$.

### 10.3 Odd depth: $x = 7/8$

For $x = 7/8$, one has $s = 3$ but still $d = 1$. Exact extraction gives

$$\mathrm{up}\left(\frac{7}{8}\right) = \frac{1}{288}, \qquad Q_n\left(\frac{7}{8}\right) = \frac{1}{288} - \frac{1}{18}4^{-n} \quad (n \geq 3). \tag{10.8}$$

The odd top derivative $\mathrm{up}^{(3)}(7/8)$ is nonzero, but only even derivatives enter the centered-tail deconvolution; hence the mode count remains $\lfloor 3/2 \rfloor = 1$.

### 10.4 Two correction modes: $x = 15/16$

Now $s = 4$ and $d = 2$. The three required finite values are

$$Q_4\left(\frac{15}{16}\right) = \frac{1}{24576}, \qquad Q_5\left(\frac{15}{16}\right) = \frac{121}{1966080}, \qquad Q_6\left(\frac{15}{16}\right) = \frac{6331}{94371840}. \tag{10.9}$$

Using (7.16),

$$\begin{aligned} \mathrm{up}\left(\frac{15}{16}\right) &= \frac{1}{45}Q_4 - \frac{4}{9}Q_5 + \frac{64}{45}Q_6 \\ &= \boxed{\frac{143}{2073600}}. \end{aligned} \tag{10.10}$$

The coefficients can be read from derivatives. At derivative order 2, $15/16$ maps to $y = 3/4$ with positive Thue–Morse sign, so

$$\mathrm{up}''\left(\frac{15}{16}\right) = 8\mathrm{up}\left(\frac{3}{4}\right) = \frac{5}{9}. \tag{10.11}$$

At derivative order 4, the point itself is a cell midpoint with positive sign, so

$$\mathrm{up}^{(4)}\left(\frac{15}{16}\right) = 2^{\binom{5}{2}} = 1024. \tag{10.12}$$

Consequently

$$C_1 = -\frac{1}{18}\frac{1}{4}\frac{5}{9} = -\frac{5}{648}, \tag{10.13}$$

$$C_2 = \frac{31}{16200}\frac{1}{16}1024 = \frac{248}{2025}. \tag{10.14}$$

The full exact law is

$$\boxed{Q_n\left(\frac{15}{16}\right) = \frac{143}{2073600} - \frac{5}{648}4^{-n} + \frac{248}{2025}4^{-2n}, \qquad n \geq 4.} \tag{10.15}$$

Cancelling only the first mode changes the observed decay from approximately $4^{-n}$ to exactly a multiple of $16^{-n}$; cancelling both returns the limit with zero residual in exact arithmetic.

### 10.5 A three-mode example

For $x = 63/64$, $s = 6$ and $d = 3$. Solving the four-by-four rational system gives

$$\begin{aligned} Q_n\left(\frac{63}{64}\right) = \frac{1153}{561842749440} - \frac{143}{18662400}4^{-n} + \frac{31}{3645}4^{-2n} \\ - \frac{4806656}{2679075}4^{-3n}, \qquad n \geq 6. \end{aligned} \tag{10.16}$$

The exact value is obtained from $Q_6, Q_7, Q_8, Q_9$ with the $d = 3$ vector (7.17). Large-looking correction coefficients are harmless because they multiply very small powers $4^{-mn}$.

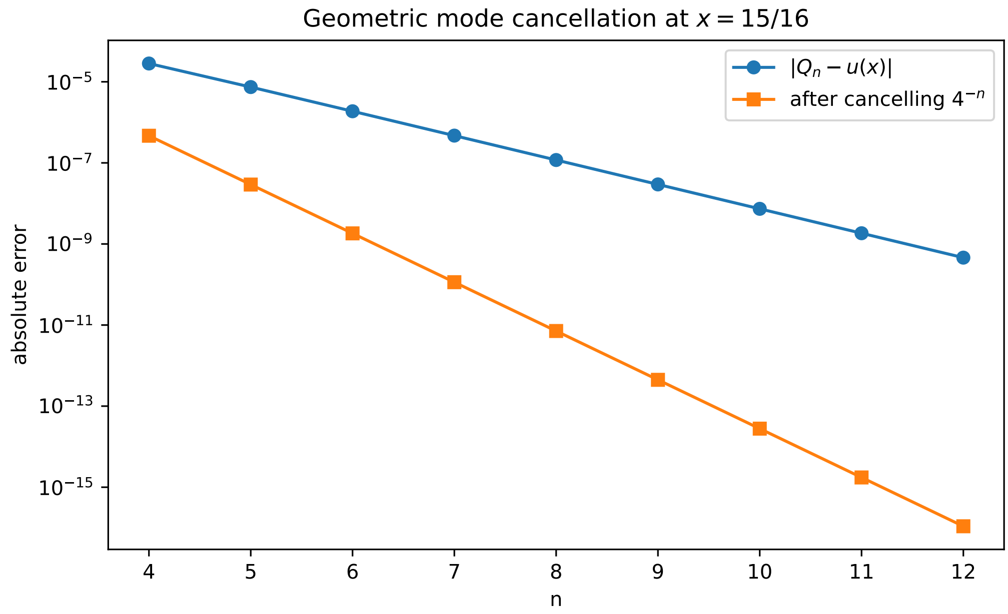


Figure 2: At $x = 15/16$, one $q$-Richardson step cancels the $4^{-n}$ mode and leaves only the $4^{-2n} = 16^{-n}$ mode. The two-mode formula cancels both exactly.

### 10.6 Symmetry

Since every finite box is even, every $p_N$ and up are even. Therefore all formulas at $x$ immediately give the corresponding formulas at $-x$:

$$Q_n(-x) = Q_n(x), \qquad \mathrm{up}(-x) = \mathrm{up}(x), \qquad C_m(-x) = C_m(x). \tag{10.17}$$

## 11 Exact extraction algorithm

**Algorithm 11.1** (Dyadic rational extraction)**.** Given a dyadic rational $x \in [-1, 1]$:

1. Reduce $x$ and determine the smallest $s$ such that $2^s x \in \mathbb{Z}$.
2. Set $d = \lfloor s/2 \rfloor$ and choose any $n_0 \geq s$; the smallest guaranteed choice is $n_0 = s$.
3. Compute the $d + 1$ exact rational spline values

$$Q_{n_0}(x), Q_{n_0+1}(x), \ldots, Q_{n_0+d}(x)$$

   using (2.8) or (2.11).
4. Compute the exact weights

$$W_{d,j} = \frac{(-1)^{d-j} \rho^{\binom{d-j+1}{2}}}{(\rho;\rho)_j (\rho;\rho)_{d-j}}, \qquad \rho = \frac{1}{4}.$$

5. Return

$$\mathrm{up}(x) = \sum_{j=0}^{d} W_{d,j} Q_{n_0+j}(x).$$

6. Optionally solve (8.3) to obtain every $C_m$, and verify the next unused value with (6.2) or the recurrence (9.2).

### 11.1 Exact arithmetic at aligned points

At a valid index $n \geq s$, let $N = n + 1$ and use the scaled coordinate $y = 2^{N-1}(x + A_N)$. Midpoint alignment makes $2y$ an odd integer. Therefore (2.11) can be evaluated without any fractional power in the summation:

$$p_N(x) = \frac{1}{2^{N(N-1)/2}(N-1)!} \sum_{0 \leq k \leq \lfloor y \rfloor} \tau_k (2y - 2k)^{N-1}. \tag{11.1}$$

This is the form used by the accompanying code. It uses only integer powers, integer Thue–Morse signs, and one final rational normalization.

### 11.2 Pseudocode

Listing 1: Core exact extraction logic.

```
s = dyadic_depth(x)
d = s // 2
rho = Fraction(1, 4)

values = [up_prefix(x, s + j) for j in range(d + 1)]
weights = []
for j in range(d + 1):
    r = d - j
    weights.append(
        (-1)**r * rho**(r*(r+1)//2)
        / (qpoch(rho, j) * qpoch(rho, r))
    )

exact_up_x = sum(w*q for w, q in zip(weights, values))
```

The complete file `verify_dyadic_up_extraction.py` includes exact finite-spline evaluation, $q$-Pochhammer and Gaussian-binomial routines, rational Gauss–Jordan elimination, analytic coefficient reconstruction from derivative cells, recurrence checks, CSV generation, and figures.

## 12 Computational verification

All numerical claims in this section were obtained with `fractions.Fraction`; no floating-point fitting was used. The script exhaustively tested every reduced dyadic point in $[-1, 1]$ of depth at most seven. There are

$$3 + \sum_{s=1}^{7} 2^s = 257 \tag{12.1}$$

such points when $-1, 0, 1$ are included at depth zero.

For each point the program performed the following independent exact checks:

1. extraction from four different valid starting windows returned the same rational value;
2. the Vandermonde-fitted coefficients reproduced six consecutive finite approximants beyond the onset;
3. the fitted coefficients agreed term by term with $(-1)^m a_m 4^{-m} \mathrm{up}^{(2m)}(x)$, with derivatives evaluated by the exact cell tiling;
4. the $q$-binomial annihilating recurrence had zero residual for six consecutive indices.

The run completed

$$\begin{array}{lr} \text{sequence and window identities} & 2570, \\ \text{analytic-versus-fitted coefficient entries} & 941, \\ \text{annihilating recurrence identities} & 1542, \end{array} \tag{12.2}$$

with every equality exact. Representative values and the complete run summary are stored in the companion CSV and text files included with the archive.

Table 1: Representative exact geometric laws $Q_n = L + \sum_{m=1}^{d} C_m 4^{-mn}$.

| $x$ | $s$ | $d$ | Exact law, valid for $n \geq s$ |
|---|---|---|---|
| $3/4$ | 2 | 1 | $Q_n = \frac{5}{72} - \frac{1}{9}4^{-n}$ |
| $7/8$ | 3 | 1 | $Q_n = \frac{1}{288} - \frac{1}{18}4^{-n}$ |
| $15/16$ | 4 | 2 | $Q_n = \frac{143}{2073600} - \frac{5}{648}4^{-n} + \frac{248}{2025}4^{-2n}$ |
| $5/16$ | 4 | 2 | $Q_n = \frac{1767743}{2073600} + \frac{67}{648}4^{-n} + \frac{248}{2025}4^{-2n}$ |
| $31/32$ | 5 | 2 | $Q_n = \frac{19}{33177600} - \frac{1}{2592}4^{-n} + \frac{124}{2025}4^{-2n}$ |

# 13 Further consequences and extensions

## 13.1 Exact rather than asymptotic Richardson acceleration

For an arbitrary point, the $M$-sample signed geometric Richardson combination cancels the first $M-1$ terms of an asymptotic expansion and leaves a smaller remainder. At a dyadic point of depth $s$, choose $M = d+1$. There is no remainder: every possible even-derivative term after order $2d$ is exactly zero. Thus the same filter changes status from an accelerator to a finite exact reconstruction operator.

This also explains why using more than $d+1$ samples is unnecessary in exact arithmetic. Extra samples are useful only for consistency checking, for averaging noisy approximate evaluations, or when the denominator depth is not known reliably.

## 13.2 A rational generating function for the eventual sequence

Let $n \geq s$ and write $R_k = Q_{s+k}(x)$. From (6.2),

$$\sum_{k=0}^{\infty} R_k z^k = \frac{\mathrm{up}(x)}{1-z} + \sum_{m=1}^{d} \frac{C_m(x)\rho^{ms}}{1-\rho^m z}. \tag{13.1}$$

Hence the tail generating function is rational, with denominator dividing

$$(1-z)\prod_{m=1}^{d}(1-\rho^m z). \tag{13.2}$$

The recurrence (9.1) is the coefficient form of this rationality.

### 13.3 Finite differences and mode peeling

The first difference removes the constant:

$$Q_{n+1} - Q_n = \sum_{m=1}^{d} C_m(\rho^m - 1)\rho^{mn}. \tag{13.3}$$

Applying $E - \rho$ removes the slowest remaining mode; applying $E - \rho^2$ next removes the second, and so on. This gives a sequential "mode-peeling" interpretation of the product filter. It is useful for symbolic diagnostics and for displaying the progressively faster slopes in a log-error plot.

### 13.4 General geometric scales

The polynomial-deconvolution lemma does not depend on the number 2. Suppose a compactly supported smooth density $u$ admits a head–tail identity

$$u = p_N * u_{\lambda^N} \tag{13.4}$$

with $0 < \lambda < 1$, and the relevant finite prefix is polynomial on a full tail neighborhood of a point. If the reciprocal Fourier symbol has only even powers, then the correction modes are

$$\lambda^{2mN} = (\lambda^2)^{mN}. \tag{13.5}$$

Whenever an independent structural theorem makes the derivative jet finite at the point, the same argument produces an exact $q$-interpolation formula with $q = \lambda^2$. In the Rvachev case, $\lambda = 1/2$ and therefore $q = 1/4$. The uniform dyadic knot alignment and the exact derivative cutoff are the special features that make the theorem especially clean here.

### 13.5 What changes away from dyadic points

At a non-dyadic point, two obstructions appear.

1. The tail interval $[x - 2^{-N}, x + 2^{-N}]$ generally crosses a knot of $p_N$, so one polynomial no longer describes the entire convolution neighborhood.
2. The derivative jet $\mathrm{up}^{(r)}(x)$ does not terminate at a finite order in general.

The all-orders expansion and $q$-Richardson acceleration remain useful, but the finite exact formula should not be expected. This contrast sharply identifies the arithmetic-geometric role of dyadic points.

## 14 Conclusions

For a dyadic point $x = a/2^s$, the finite sinc-product splines contain the exact value $\mathrm{up}(x)$ in a surprisingly small amount of data. After index $n = s$ the entire sequence is a degree-$\lfloor s/2 \rfloor$ polynomial in $4^{-n}$. The mechanism is exact at every stage:

1. the finite Fourier product is a uniform-knot Thue–Morse spline;
2. the point becomes the midpoint of a cell once $n \geq s$ in the user indexing;
3. the omitted tail is supported precisely inside that cell;
4. reciprocal-sinc deconvolution is finite on the cell polynomial;
5. the derivative tiling kills all orders above $s$;
6. a $q$-Pochhammer filter annihilates the remaining geometric modes.

The final reconstruction is the single formula

$$\operatorname{up}(x) = \sum_{j=0}^{d} \frac{(-1)^{d-j} 4^{-\binom{d-j+1}{2}}}{(1/4;1/4)_j (1/4;1/4)_{d-j}} Q_{n_0+j}(x), \qquad d = \lfloor s/2 \rfloor, \; n_0 \geq s,$$

or, equivalently, the integer-coefficient base-4 Gaussian-binomial formula (7.12). The Vandermonde interpolation polynomial recovers all correction coefficients as well. In exact arithmetic this is a finite algorithm for the rational dyadic values of Rvachev's up-function; in numerical arithmetic it is also stable, with weight amplification uniformly below 1.97.

# A Proof of the Lagrange weight formula

For completeness, evaluate the Lagrange basis (8.5) at zero. The numerator is

$$\prod_{\substack{0 \leq r \leq d \\ r \neq j}} (-\rho^r) = (-1)^d \rho^{d(d+1)/2-j}. \tag{A.1}$$

For $r < j$,

$$\rho^j - \rho^r = -\rho^r(1-\rho^{j-r}),$$

so those factors contribute

$$(-1)^j \rho^{j(j-1)/2} (\rho;\rho)_j.$$

For $r > j$,

$$\rho^j - \rho^r = \rho^j(1-\rho^{r-j}),$$

so those factors contribute

$$\rho^{j(d-j)} (\rho;\rho)_{d-j}.$$

Their quotient is

$$\begin{aligned} \ell_j(0) &= \frac{(-1)^{d-j} \rho^{d(d+1)/2-j-j(j-1)/2-j(d-j)}}{(\rho;\rho)_j (\rho;\rho)_{d-j}} \\ &= \frac{(-1)^{d-j} \rho^{(d-j)(d-j+1)/2}}{(\rho;\rho)_j (\rho;\rho)_{d-j}}, \end{aligned} \tag{A.2}$$

which is exactly (7.8).

# B Finite $q$-binomial identities used in the article

The finite $q$-binomial theorem is

$$(a;q)_d = \sum_{r=0}^{d} (-1)^r q^{\binom{r}{2}} \begin{bmatrix} d \\ r \end{bmatrix}_q a^r. \tag{B.1}$$

With $a = q/z$, multiplication by $z^d$ yields

$$\begin{aligned} \prod_{m=1}^{d} (z - q^m) &= z^d (q/z;q)_d \\ &= \sum_{r=0}^{d} (-1)^r q^{r(r+1)/2} \begin{bmatrix} d \\ r \end{bmatrix}_q z^{d-r}. \end{aligned} \tag{B.2}$$

Putting $j = d - r$ gives (7.5). Replacing $a$ by $-q$ in (B.1) gives

$$\prod_{m=1}^{d}(1+q^m) = \sum_{r=0}^{d} q^{r(r+1)/2}\begin{bmatrix} d \\ r \end{bmatrix}_q, \tag{B.3}$$

which proves the variation formula (7.19).

# C Reproducibility files

The archive accompanying this article contains:

`dyadic_up_extraction.tex`
: The complete LaTeX source.

`dyadic_up_extraction.pdf`
: The rendered article.

`verify_dyadic_up_extraction.py`
: A documented exact-arithmetic implementation and verification suite.

`sample_sequences.csv`
: Representative finite values, exact limits, and errors.

`geometric_coefficients.csv`
: Fitted exact coefficients for representative points.

`verification_report.txt`
: The exhaustive depth-seven verification summary and initial coefficient tables.

`finite_splines.pdf/.png`
: The generated finite-spline figure.

`geometric_mode_cancellation.pdf/.png`
: The generated mode-cancellation figure.

To reproduce the tables and figures, run

```
python verify_dyadic_up_extraction.py --out-dir . --max-depth 7
```

The exact verification can be run without plotting dependencies by adding `–skip-plots`.

# References


[1] V. Reshetnikov, *The Fabius Function and Rvachev's Up-Function: Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics*, `ProveIt` repository documentation, https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up.

[2] V. Reshetnikov, *Exponents and $q$-Series Frontiers*, especially the finite sinc-product spline, head–tail, all-orders, and signed geometric Richardson sections, `ProveIt` repository documentation, https://github.com/VladimirReshetnikov/ProveIt/tree/main/Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/exponents-and-q-series.

[3] J. Arias de Reyna, *An infinitely differentiable function with compact support: Definition and properties*, English translation of the 1982 paper, arXiv:1702.05442 (2017).

[4] J. Arias de Reyna, *Arithmetic of the Fabius function*, arXiv:1702.06487 (2017).

[5] NIST Digital Library of Mathematical Functions, *Chapter 17: q-Hypergeometric and Related Functions*, especially Section 17.2, `https://dlmf.nist.gov/17.2`.

[6] G. Gasper and M. Rahman, *Basic Hypergeometric Series*, 2nd ed., Encyclopedia of Mathematics and its Applications 96, Cambridge University Press, 2004.